Review: Riemann Sums, Definite Integrals & the Fundamental Theorem of Calculus

MATH 146 – Calculus II

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Riemann Sums

Consider a function $f(x)$ defined on an interval $a \leq t \leq b$, and, given a whole number n, divide the interval into n subintervals of length $\Delta t = \dfrac{(b-a)}{n}$ with endpoints $t_0, t_1, t_2, \ldots, , t_n$.

Left Riemann Sum

The **left (regular) Riemann sum** is:

$$f(t_0)\Delta t + f(t_1)\Delta t + \cdots + f(t_{n-1})\Delta t = \sum_{i=0}^{n-1} f(t_i)\Delta t.$$

Right Riemann Sum

The **right (regular) Riemann sum** is:

$$f(t_1)\Delta t + f(t_2)\Delta t + \cdots + f(t_n)\Delta t = \sum_{i=1}^{n} f(t_i)\Delta t.$$

Definite Integral

Definite Integral

Given a continuous function $f(x)$ on an interval $a \leq t \leq b$, the **definite integral** of $f(x)$ over $[a, b]$ is the limit:

$$\int_a^b f(t)\,dt = \lim_{n\to\infty} \sum_{i=1}^{n} f(t_i)\Delta t,$$

or:

$$\int_a^b f(t)\,dt = \lim_{n\to\infty} \sum_{i=0}^{n-1} f(t_i)\Delta t$$

provided these limits exist. $f(x)$ is called the **integrand** and a and b are called the **limits of integration**.

Example: Definite Integral

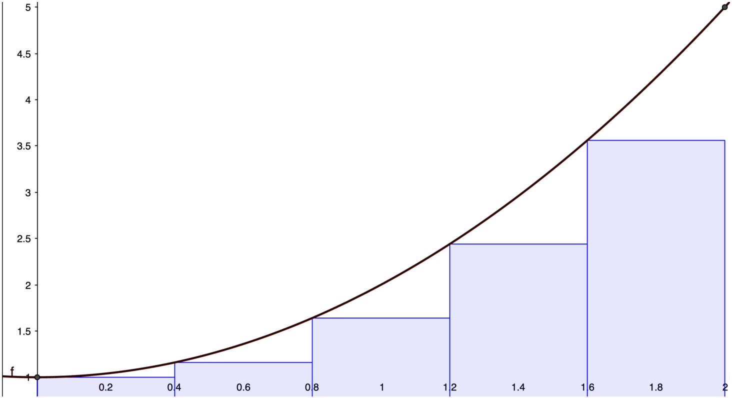
Example: Consider the function $f(x) = x^2 + 1$ over the interval $[0, 2]$. For a given n, $\Delta t = {}^2/_n$ and $t_i = {}^{2i}/_n$, so that:

$$\int_0^2 x^2 + 1\,dx = \lim_{n\to\infty} (({}^{2i}/_n)^2 + 1) \times ({}^2/_n).$$

This sum can be thought of as a sum of areas of rectangles "under" the graph of $f(x)$.

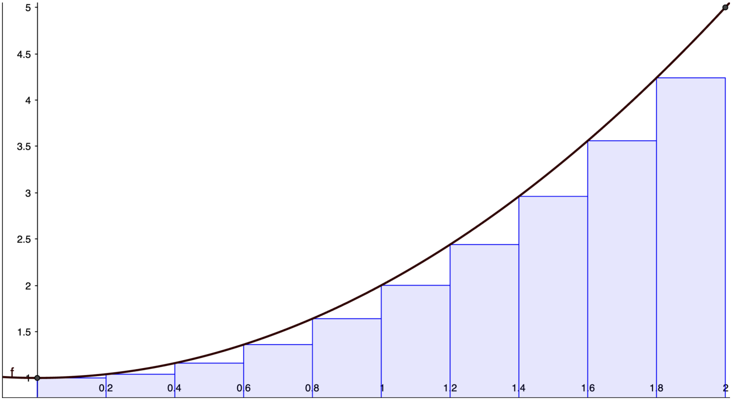
Example: $n = 5$

$$\int_0^2 x^2 + 1\,dx = \sum_{i=0}^{4} \left(\left({}^{2i}/_{5} \right)^2 + 1 \right) \times \left({}^{2}/_{5} \right) = 4.384$$



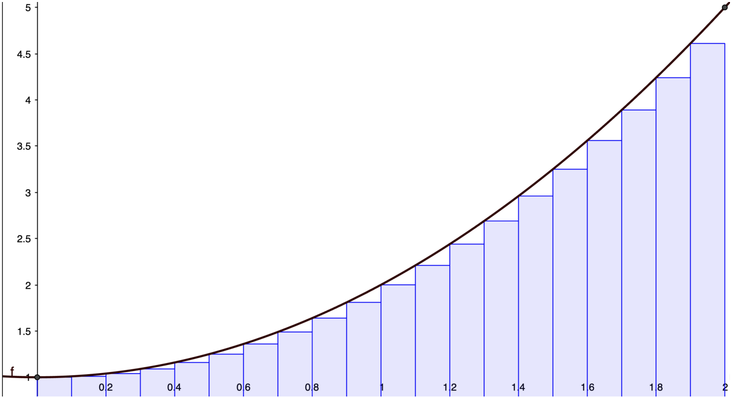
Example: $n = 10$

$$\int_0^2 x^2 + 1\,dx = \sum_{i=0}^{9} \left(\left({}^{2i}\!/\!_{10} \right)^2 + 1 \right) \times \left({}^{2}\!/\!_{10} \right) = 4.448$$



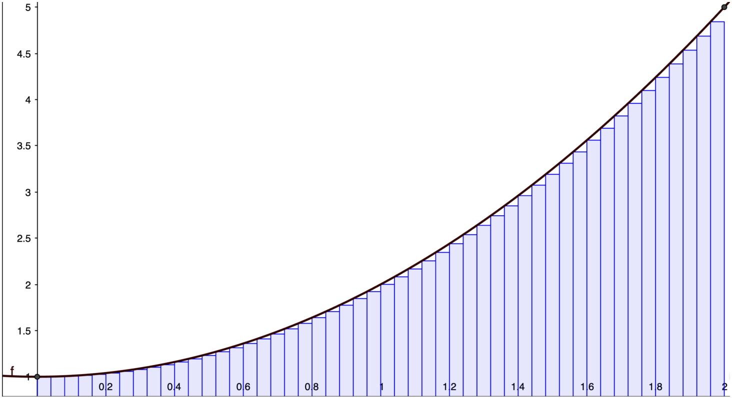
Example: $n = 20$

$$\int_0^2 x^2 + 1\,dx = \sum_{i=0}^{19} \left(\left({}^{2i}\!/_{20} \right)^2 + 1 \right) \times \left({}^{2}\!/_{20} \right) = 4.571$$



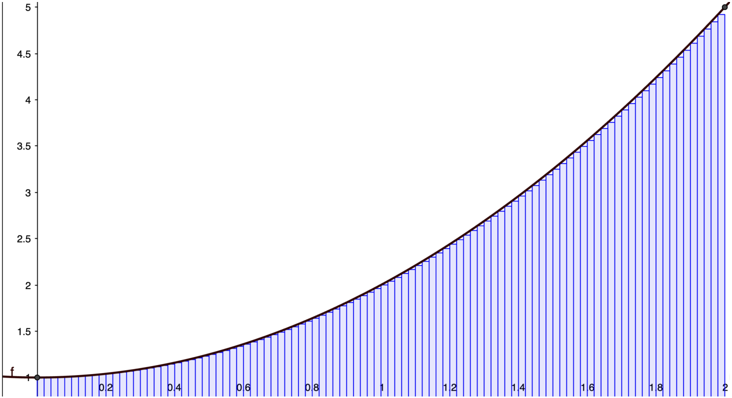
Example: $n = 50$

$$\int_0^2 x^2 + 1\,dx = \sum_{i=0}^{49} \left(\left({}^{2i}\!/_{50} \right)^2 + 1 \right) \times \left({}^{2}\!/_{50} \right) = 4.627264$$



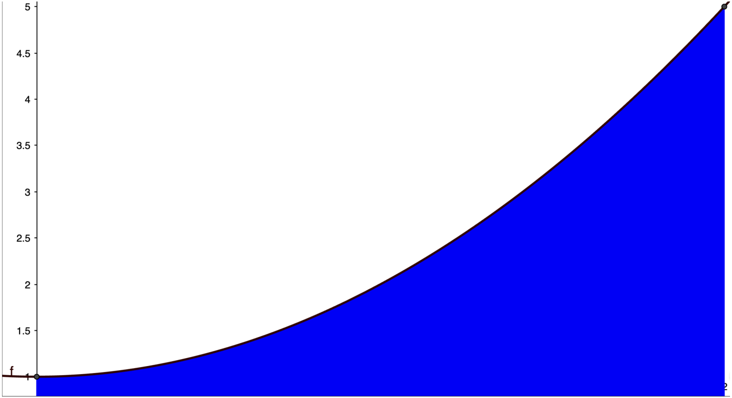
Example: $n = 100$

$$\int_0^2 x^2 + 1\,dx = \sum_{i=0}^{99} \left(\left({}^{2i}/_{100} \right)^2 + 1 \right) \times \left({}^{2}/_{100} \right) = 4.646808$$



Example: $n = 1000$

$$\int_0^2 x^2 + 1\, dx = \sum_{i=0}^{999} \left(\left({}^{2i}\!/\!_{1000} \right)^2 + 1 \right) \times \left({}^{2}\!/\!_{1000} \right) = 4.664668$$



Example: $n \to \infty$

As $n \to \infty$, this sum becomes the *area under the graph* of $f(x)$ over $[a, b]$.

$$\int_0^2 x^2 + 1\, dx = \lim_{i\to\infty} \sum_{i=0}^{n-1} ((2i/n)^2 + 1) \times (2/n) = 14/3$$

Estimates of $\int_a^b f(x)\, dx$ can be obtained with left sums right sums, sums from midpoint values of $f(x)$ or more general types of sums.

In simple cases, the values of $\int_a^b f(x)\, dx$ can be found simply by finding an area under a graph.

Example: Definite Integral

Example: Evaluate $\int_0^2 -2x + 1\,dx$.

$\int_a^b f(x)\,dx$ is the *signed* area – that is, the area between the graph of $f(x)$ and the x-axis.

Velocity and Distance

Recall the definition of rate of change: $r = \dfrac{d}{t}$.

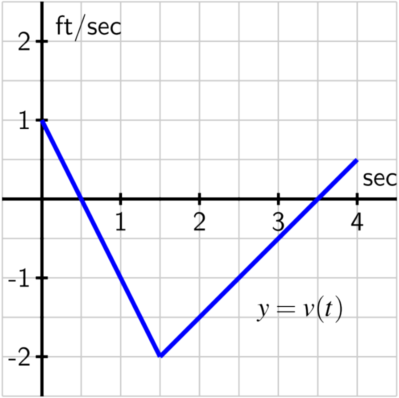
This implies that $d = rt$.

Suppose $v(t)$ is the velocity of an object of an object at time t and consider the distance traveled over the time interval Δt.

$$\begin{aligned} \int_a^b v(t)\,dt &= \lim_{n\to\infty} \sum_{i=1}^{n} v(t_i)\Delta t \\ &= \text{distance traveled (displacement)} \\ &= \text{area under the graph of } v(t) \end{aligned}$$

Example: Velocity and Distance

The velocity of an object moving along an axis is described by the picture below. Find the displacement of the object from time $t = 0$ to time $t = 4$.



The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus

Suppose $f(t)$ is a continuous function on the interval $[a, b]$ and that $F(t)$ is an antiderivative of $f(t)$. Then

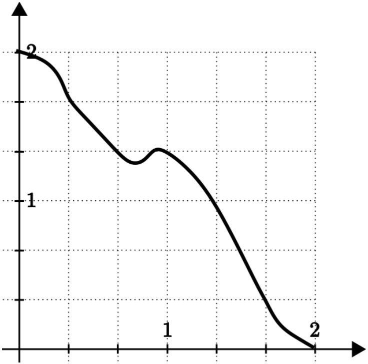
$$\int_a^b f(t)\,dt = F(b) - F(a)$$

Notes:

- This makes evaluating definite integrals easy, *provided you can find an antiderivative.*
- Alternatively, $\displaystyle\int_a^b f'(t)\,dt = f(b) - f(a)$.
- Also, $F(b) = F(a) + \displaystyle\int_a^b f(t)\,dt$.

Example: The Fundamental Theorem of Calculus

Example: At 10 a.m. A cistern at a rate of $R(t)\,{}^{gallons}/_{hour}$, where $R(t)$ has the graph given below.



Estimate the amount of water in the tank at noon.

Example: The Fundamental Theorem of Calculus

Examples:

- Find $\displaystyle\int_0^2 3x + 1\, dx$.

- Find $\displaystyle\int_1^3 (x-1)^2\, dx$.

Example: The Fundamental Theorem of Calculus

Examples:

- Find $\displaystyle\int_0^{\frac{\pi}{2}} \sin(x)\,dx$.

- Find $\displaystyle\int_0^{3} e^x\,dx$.

- Find $\displaystyle\int_0^{\pi} \cos(x) + \frac{1}{\sqrt{1-x^2}}\,dx$.

Some Theorems for Definite Integrals

- $\int_a^b c \cdot f(x) + d \cdot g(x)\,dx = c\int_a^b f(x)\,dx + d\int_a^b g(x)\,dx.$

- If $a < c < b$, $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$

Note: There is no product or quotient rule for definite integrals.

Every continuous function $f(x)$ has an antiderivative.

The Fundamental Theorem of Calculus

If $f(x)$ is a continuous function on an interval and a is any point, then the function:

$$F(x) = \int_a^x f(t)\, dt,$$

is an antiderivative of $f(x)$. In other words:

$$\frac{d}{dx}\left[\int_a^x f(t)\, dt\right] = f(x).$$